INCH-POUND

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SUPERSEDING
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# COMMERCIAL ITEM DESCRIPTION

LOAD BANK, ELECTRICAL, 160 Kilowatts, 60 Hertz

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID describes an alternating current (AC), 160 kilowatt (kW), 60 Hertz (Hz), portable type, resistive load bank and accessories. The commercially-available, portable, enclosed, forced air-cooled, industrial load bank is used for electrical performance testing of tactical power generators and other 60 Hz generator and power supply systems used for ground-based power operations.

2. SALIENT CHARACTERISTICS.

2.1 Safety and environmental. The equipment shall meet all safety and environmental requirements as specified in National Fire Protection Association (NFPA) 70 – National Electrical Code (NEC), American National Standards Institute (ANSI) IEEE C2 – National Electrical Safety Code, and the latest applicable National Electrical Manufacturer's Association (NEMA) standards, except as stated herein. All resistive load element buses shall be over-current protected. Airflow failure shall cause the load bank to automatically drop all connected load before an over-temperature condition occurs.

2.2 Electrical power sources. The load bank shall be completely self-contained to include all controls, connection points, instrumentation, load elements, and system protective devices. The operating power for internal circuits, instruments, and fan(s) shall be user-selectable between being derived only from the power source under test, or from an external power source. All controls shall be located on a local control panel.

2.3 Operating temperature range. The equipment shall meet its performance and accuracy requirements in an operating environment of -20 to 125 $^{o}$F.

2.4 Non-operating temperature. The equipment shall meet its performance and accuracy requirements (see 2.8.1.1) after being in a non-operating environment (storage temperature) of -40 to 160 $^{o}$F.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: WR-AFLCMC/WNZEA, 235 Byron St., Suite 19A, Robins AFB GA 31098-1813. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

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2.5 Calibration and maintenance adjustments. The design of the system shall provide for readily-accessible calibration adjustments and maintenance adjustments. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. These adjustments shall be provided by variable value components. The calibration by substitution of selected components or parts is unacceptable, unless specifically approved. The calibration interval shall be a period of one year or greater based on an operating time of 2,000 hours.

2.6 Accessories. Accessories to be supplied by the manufacturer with each deliverable load bank shall include standard and optional accessories as provided by the manufacturer, and listed in the manufacturer's catalog or data sheet. A written user's manual, a written programmer-reference manual (if applicable), and a traceable calibration certificate shall also be provided with each load bank unit. The user manual(s) shall include complete details for the operation, troubleshooting, and maintenance of the load bank, and shall include interconnect drawings and electrical schematics for all power and control circuits and devices. A complete parts listing shall be delivered with each load bank, and shall include, at a minimum, the original manufacturer's name, part number, rating, identification, and quantity required.

2.7 Environmental requirements. The load bank shall be designed for indoor or sheltered use and indoor storage. The load bank shall be clearly marked, "NOT FOR USE IN HAZARDOUS LOCATIONS."

2.8 Performance characteristics. The load bank shall be capable of continuous, resistive load testing of ground-power generator sets by providing loads ranging from 0 to not less than 160 kW load at 240 and 480 volts AC, 3-phase, 4-wire, 60 Hz. The load bank shall also be capable of providing loads ranging from 0 to not less than 120 kW at 208 and 416 volts AC, 3-phase, 4-wire, 60 Hz.

2.8.1 Controls. The load bank shall include a local operator control panel containing all meters, switches, alarms, indicators, and instructions necessary for the proper operation of the load bank. A cover assembly shall be provided to protect the control panel from damage when not in use.

2.8.1.1 Meters. The control panel of the load bank shall be equipped with metering for monitoring voltage (0-600 volts AC), frequency (45-65 Hz), and current (0-400 amps AC) for all phases. Meter accuracy shall be within ±3%, full scale.

2.8.1.2 Phase selector switch. The load bank shall have a phase selector switch or other means on the control panel for viewing the current of each phase of incoming power (A, B, and C). Voltage measurements shall be viewable line-to-line for each phase of incoming power.

2.8.2 AC input connections. The load bank input shall be equipped with portable power cables rated for extra-hard usage and capable of continuously carrying the load bank's rated full-load current. Input-power cable length shall not be less than 15 feet per phase conductor with terminal lugs on the ends for connection to power supply studs. External control power or cooling-fan power input cable shall be designed for 120 volts AC, 60 Hz, single phase, be provided with a properly rated input connector, be rated for extra-hard usage, and not be less than 15 feet in length. All input cables shall be highly-flexible copper conductor and rated for 600 volts minimum. Input-power cable storage shall be provided on the load bank.

2.8.3 Load elements. The load bank load elements shall be of the purely resistive type, and shall be designed so that a load of any magnitude within the range of 1 to 160 kW may be absorbed.



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Load elements shall be protected from damage per NFPA 70, article 110.10, by the use of user-resettable circuit breakers, user-replaceable fuses, or automatically-resetting thermal switches. All load elements shall be connectable to incoming power by means of electrical contactors. Load balance shall be within ±2.5% between phases.

2.8.4 Step loading. The load bank shall provide a means to apply incremental loading (step loading). The following load steps should be available, at a minimum: 1 kW, 2 kW, 5 kW, 10 kW, 20 kW, and 50 kW. The use of multiple load steps of the same size is acceptable. Load steps of a different resolution may be acceptable, but must be identified at time of quote/proposal submission. Government review will be accomplished and approval or disapproval provided. The tolerance of each load step shall be no more than 5%. These loads shall be selectable individually or in combinations to provide loads up to and including 160 kW.

2.8.5 Shock loading. The load bank shall have a single, master load switch for shock loading the unit under test at any load setting described in paragraph 2.8.4. The single switch may also serve as a master load disconnect switch.

2.8.6 Grounding. Electrical grounding shall be accomplished in accordance with NFPA 70, article 250. An external ground terminal shall be provided so that a separate external user ground-conductor connection may be added. This ground terminal shall be bonded to the main metal frame and chassis of the load bank, be clearly marked, easily accessible, and within sight of the operator when standing at the control panel.

2.8.7 Over-temperature. The load bank shall contain adequate protection circuitry to automatically protect the load elements and controls, and shall immediately cease operation in the event of a cooling failure or over-temperature condition.

2.8.8 Over-voltage. The load bank shall contain adequate protection circuitry to automatically cease operation to protect the load elements, controls, and operator in the event of an over-voltage condition on the input power.

2.8.9 Load disconnect. The load bank shall have provision for the manual de-activation of load elements in case of a failure of the automatic protection system. The load disconnect shall provide a single-point for instantaneous load removal from the load bank.

2.8.10 Continuous operation. The load bank shall be capable of a continuous run at full load for up to 4 hours. The load bank shall be capable of a cool down to ambient temperature within 5 minutes with cooling fan on, under no-load conditions, and be capable of performing multiple back-to-back full loads without damage to components. The load bank shall be capable of withstanding instantaneous input-power disconnection with cooling-fan stoppage at full load without damage to the load bank components.

2.9 Construction. The load bank shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, service, and operation. Electrical components shall be designed to withstand transient voltage- and current-level sags and surges associated with load testing, stand-alone generator sets. Enclosure finishes and other components shall employ best commercial-design practices, and be designed to withstand the high temperatures and exposures normally associated with load bank operations. Cooling-air intake and exhaust openings shall be covered with screens or louvers to protect load elements from foreign object



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damage. Access to cooling fan(s), instrumentation wiring, contactors, load elements, protective devices, and associated wiring shall be through removable panels or hinged doors/access panels.

2.9.1 Transportability. The load bank shall be provided with wheels having a diameter of 5 ±1 inches to allow easy movement over hard surfaces, and with lockable casters on at least two wheels on the same side of the load bank. The load bank shall be constructed to have the capability of being lifted and transported with a forklift by means of forklift tine guides incorporated into the load bank enclosure. The load bank shall have provisions for sling-lifting or eye-lifting without sustaining damage.

2.9.2 Size. The load bank overall dimensions shall not exceed 60 inches long by 36 inches wide by 50 inches high, including handles and wheels.

2.9.3 Weight. The weight of the load bank shall not exceed 700 pounds (not including input power cables).

2.9.4 Fire prevention and safety. The enclosure of the load bank shall employ best commercial-design practices to reduce the danger to personnel from fire and explosion. Suitable vents and drains shall be incorporated into the enclosure of the load bank to reduce the risk of hazards.

2.9.5 Workmanship and wiring. The load bank shall be manufactured with the best commercial practices and be free from defects. All wiring shall be neatly routed, terminated, and identified.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or biobased materials. The contractor is encouraged to use recovered materials or biobased products to the maximum extent practicable, in accordance with Federal Acquisition Regulation (FAR), part 23.403.

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100 percent of EPA-designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meeting reasonable performance standards; or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

4. PRODUCT CONFORMANCE.

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The Government reserves the right to require proof of such conformance.

4.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance. Proof of conformance may



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include, but shall not be limited to, the performance of operational tests, lab tests, modeling and simulation, and delivery of reports and data from these tests.

4.3 Responsibility of inspection. Unless otherwise specified in the contract or CID, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or CID, the contractor may use his or her own facilities, or any other facilities, suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.4 Examination of product. Each load bank shall be examined to verify compliance with the requirements herein prior to accomplishing any other tests listed in 4.5. Particular attention shall be given to materials, workmanship, dimensions, weight, proper operation of protective devices, proper operation of user controls, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings.

4.5 Operational test. After complete assembly and prior to shipment, each load bank shall be operated at rated full load for not less than 10 minutes using each of the input-voltage configurations specified in 2.8 herein to verify that all parts and controls are functional, and are functioning according to the requirements of this CID.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 American National Standards Institute (ANSI). Copies of the National Electrical Safety Code (IEEE C2) standard may be obtained from ANSI, 25 West 43rd Street, 4th floor, New York, NY 10036, or online at https://www.ansi.org/.

6.1.2 Code of Federal Regulations (CFR). CFRs may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402, or online at http://www.ecfr.gov/cgi-bin/ECFR?page=browse.

6.1.3 Electronic Components Industry Association (ECIA). Copies of these documents are available from ECIA, 111 Alderman Drive, Suite 400, Alpharetta, GA 30005, or online at https://www.ecianow.org/.

6.1.4 Federal Acquisition Regulations (FAR). FARs may be obtained from the Superintendent of Documents, Government Printing Office, Washington, DC 20402, or online at https://www.acquisition.gov/?q=browsefar.

6.1.5 Government documents. Copies of specifications, standards, and handbooks are available from DLA Document Services, 700 Robbins Avenue, Building 4/D, Philadelphia, PA 19111-5094, or online at https://assist.dla.mil.

6.1.6 National Electrical Manufacturers Association (NEMA). Copies of these documents may be obtained from NEMA, 1300 North 17th Street, Suite 900, Arlington, VA 22209, or online at http://www.nema.org/pages/default.aspx.



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6.1.7 National Fire Protection Association (NFPA). Copies of these documents may be obtained from NFPA, 1 Batterymarch Park, Quincy, MA 02169-7471, or online at http://www.nfpa.org/.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.

b. Packaging requirements (see 5).

c. Product conformance provisions.

d. Product literature.

6.3 Key words.

Generator
Ground power supplies
Portable type



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MILITARY INTEREST

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Army – AV

Preparing Activity:
Air Force – 184

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